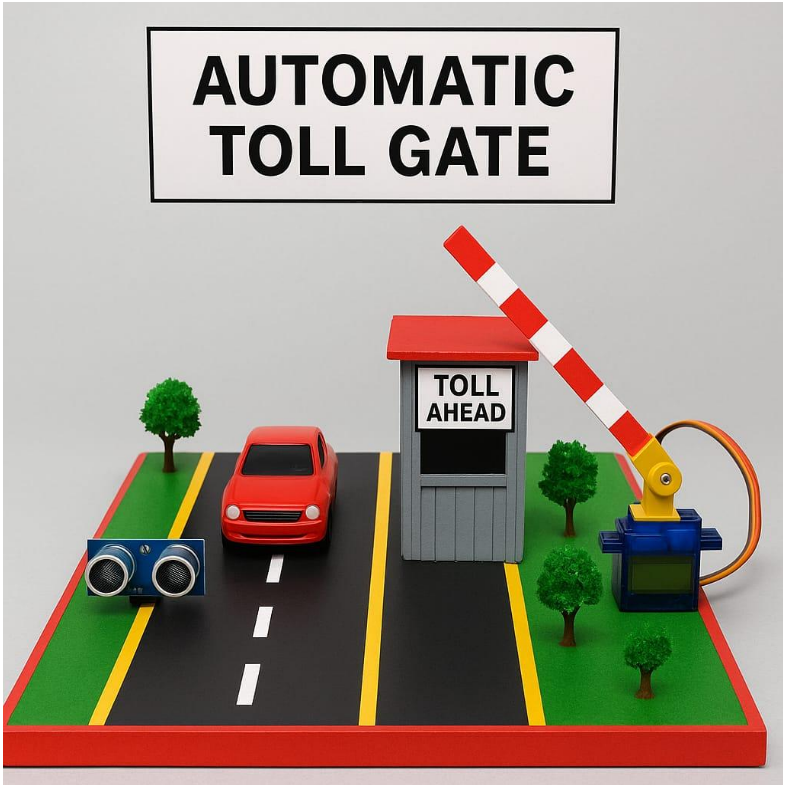

AUTOMATIC TOLL GATE SYSTEM

MAKING LIVES BETTER

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SEMESTER PROJECT- ELECTRICAL ENGINEERING

<u>SEMESTER PROJECT</u>

APPARATUS:

- Servomotor (180 degree)
- Arduino UNO
- Jumper wires
- Breadboard
- Ultra-sonic Sensor

For assembling:

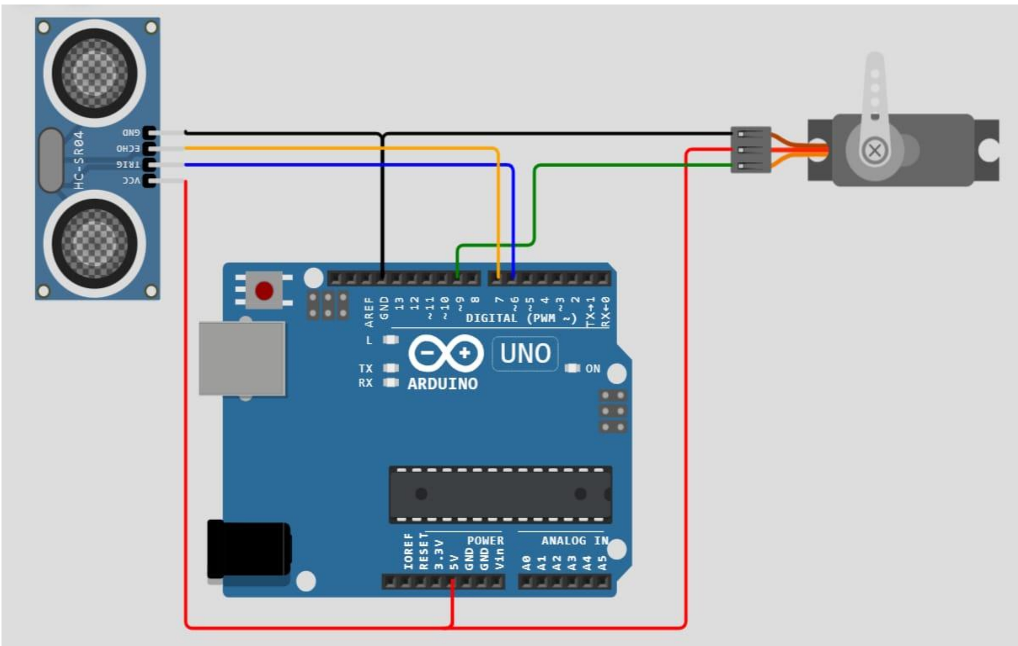
- A4 coloured papers
- Acrylic sheets
- Glue Gun

WORKING:

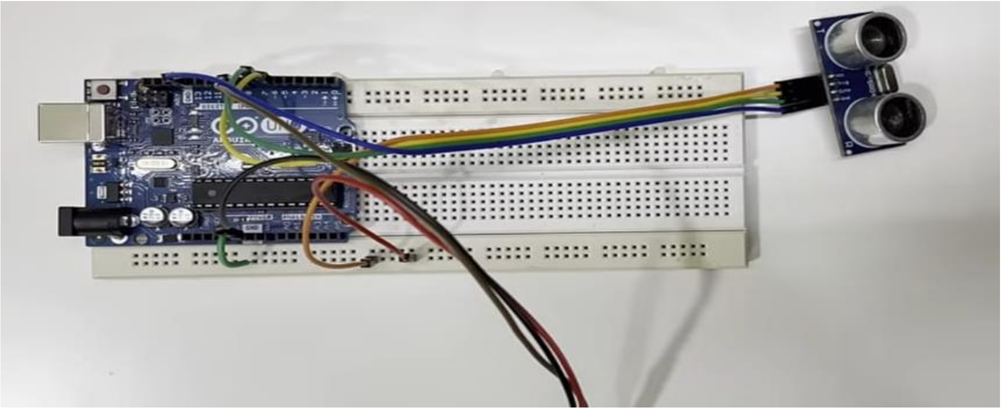
The project is an Automatic Toll Gate System designed to simulate how a car is detected and the gate opens automatically. When a vehicle approaches the toll, the ultrasonic sensor detects its presence and sends a signal to the Arduino. The Arduino processes this input and activates a servo motor, which rotates to 90 degrees to open the gate. Once the car passes and is no longer detected, the servo returns to 0 degrees, closing the gate. This setup demonstrates how toll gates or parking barriers work automatically without human involvement, using basic electronics and automation. This demonstrates a real-life toll booth or parking entry system where the gate operates automatically based on car detection, without any manual input.

CIRCUIT/ARDUINO DIAGRAMS:

CIRCUIT DIAGRAM:



CONNECTIONS:



CODE:

```
#include <Servo.h>

#define SENSOR 7    // define pin 7 for sensor
#define ACTION 6    // define pin 6 as for ACTION (relay or other output)
#define SERVO_PIN 9 // define pin 9 for servo motor

Servo myServo;

void setup() {
  Serial.begin(9600); // setup Serial Monitor to display information
  pinMode(SENSOR, INPUT_PULLUP); // define pin as Input for sensor
  pinMode(ACTION, OUTPUT); // define pin as OUTPUT for ACTION (relay)

  myServo.attach(SERVO_PIN); // attach the servo motor to pin 9
  myServo.write(0); // initial servo position (e.g., 0 degrees)
}

void loop() {
  int L = digitalRead(SENSOR); // read the sensor

  if (L == 0) {
    Serial.println("All Clear");
    digitalWrite(ACTION, LOW); // turn off relay or action pin
    myServo.write(0); // move servo to 0 degrees (safe position)
  } else {
    Serial.println("=== Obstacle detected");
    digitalWrite(ACTION, HIGH); // turn on relay or action pin
    myServo.write(90); // move servo to 90 degrees (blocked or stop position)
  }
```

```
  delay(100);
}
```

INSIGHTS

This Arduino code is used in an automatic car parking system to control a gate using a sensor and a servo motor. The sensor, connected to pin 7, detects whether a car is present. When no car is detected (sensor reads LOW), the system prints “All Clear” on the Serial Monitor, keeps the gate closed by setting the servo to 0 degrees, and turns off the relay connected to pin 6. However, when a car is detected (sensor reads HIGH), it prints “Obstacle detected,” turns on the relay (which could trigger a light or buzzer), and rotates the servo to 90 degrees to open the gate. This setup ensures the gate only opens when a car is detected, making parking access automatic and efficient. The system checks for updates every 100 milliseconds to respond quickly to changes.